# lsqvce

## Least-squares adjustment with variance component estimation

The mathematical framework

**Scope**

This document states what the toolbox computes and why: the functional model and its solution, the stochastic model, the estimation of variance components, robust and nonlinear estimation, and the quality measures. How to use the toolbox is the subject of the Manual (`doc lsqvce`); the interfaces are specified in the API reference. Equation numbers of the form "Ellmer (6.5.x)" refer to Ellmer (2018).

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# 1 The linear model and its solution

## 1.1 Gauss–Markov model

The observations $\boldsymbol{l} \in \mathbb{R}^n$ are modelled as

$$\boldsymbol{l} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{e}, \qquad \mathrm{E}\{\boldsymbol{e}\} = \boldsymbol{0}, \qquad \mathrm{C}\{\boldsymbol{e}\} = \sigma_0^2 \boldsymbol{\Sigma}, \tag{1.1}$$

with the design matrix $\boldsymbol{A}$ ($n \times u$, full column rank unless stated otherwise), the unknown parameters $\boldsymbol{x}$, the cofactor matrix $\boldsymbol{\Sigma}$ and the variance of unit weight $\sigma_0^2$. The best linear unbiased estimate and its covariance are

$$\hat{\boldsymbol{x}} = \boldsymbol{N}^{-1}\boldsymbol{n}, \quad \boldsymbol{N} = \boldsymbol{A}^\mathsf{T}\boldsymbol{P}\boldsymbol{A}, \quad \boldsymbol{n} = \boldsymbol{A}^\mathsf{T}\boldsymbol{P}\boldsymbol{l}, \quad \boldsymbol{P} = \boldsymbol{\Sigma}^{-1}, \qquad \hat{\sigma}_0^2 = \frac{\hat{\boldsymbol{v}}^\mathsf{T}\boldsymbol{P}\hat{\boldsymbol{v}}}{n-u}, \quad \boldsymbol{\Sigma}_{\hat{\mathrm{x}}\hat{\mathrm{x}}} = \hat{\sigma}_0^2 \boldsymbol{N}^{-1}, \tag{1.2}$$

with the residuals $\hat{\boldsymbol{v}} = \boldsymbol{l} - \boldsymbol{A}\hat{\boldsymbol{x}}$. The residual is a projection, $\hat{\boldsymbol{v}} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{l}$ with the hat matrix $\boldsymbol{H} = \boldsymbol{A}\boldsymbol{N}^{-1}\boldsymbol{A}^\mathsf{T}\boldsymbol{P}$, and the redundancy $n - u = \mathrm{tr}(\boldsymbol{I} - \boldsymbol{H})$ is distributed over the observations by the diagonal of $\boldsymbol{R} = \boldsymbol{I} - \boldsymbol{H}$, $\sum_i r_i = n - u$. Every variance estimator of Chapter 3 is a specialisation of "quadratic form over its expectation".

## 1.2 Whitening

With a factor $\boldsymbol{W}$ of $\boldsymbol{P} = \boldsymbol{W}^\mathsf{T}\boldsymbol{W}$ the model becomes $\bar{\boldsymbol{l}} = \bar{\boldsymbol{A}}\boldsymbol{x} + \bar{\boldsymbol{e}}$ with $\bar{\boldsymbol{A}} = \boldsymbol{W}\boldsymbol{A}$, $\bar{\boldsymbol{l}} = \boldsymbol{W}\boldsymbol{l}$ and $\mathrm{C}\{\bar{\boldsymbol{e}}\} = \sigma_0^2 \boldsymbol{I}$, and (1.2) is ordinary least squares. The toolbox takes $\boldsymbol{W} = \boldsymbol{L}^{-1}$, the inverse of the lower Cholesky factor of $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^\mathsf{T}$. $\boldsymbol{\Sigma}$ is never formed for all observations: observations are grouped into *arcs*, independent of each other, and every arc is whitened by its own law (Chapter 2). All quantities below — normal equations, residuals, redundancies, test statistics — are expressed in the whitened frame with a-priori $\sigma_0 = 1$; "residual" means $\bar{\boldsymbol{v}} = \boldsymbol{W}\hat{\boldsymbol{v}}$ unless said otherwise.

## 1.3 Two ways to accumulate and solve

**Square-root information (QR, the default).** The upper-triangular factor $\boldsymbol{R}$ of $[\bar{\boldsymbol{A}}\ \bar{\boldsymbol{l}}]$ is updated batch by batch by Householder reduction, $\boldsymbol{R} \leftarrow \mathrm{triu\,qr}\big([\boldsymbol{R};\ \bar{\boldsymbol{A}}_k\ \bar{\boldsymbol{l}}_k]\big)$, and truncated to $u+1$ rows. With

$$\boldsymbol{R} = \begin{bmatrix} \boldsymbol{R}_1 & \boldsymbol{r} \\ \boldsymbol{0}^\mathsf{T} & \rho \end{bmatrix}, \qquad \hat{\boldsymbol{x}} = \boldsymbol{R}_1^{-1}\boldsymbol{r}, \qquad \bar{\boldsymbol{v}}^\mathsf{T}\bar{\boldsymbol{v}} = \rho^2, \qquad \boldsymbol{\Sigma}_{\hat{\mathrm{x}}\hat{\mathrm{x}}} = \hat{\sigma}_0^2 (\boldsymbol{R}_1^\mathsf{T}\boldsymbol{R}_1)^{-1},$$

so the residual sum of squares is obtained without subtraction, and the condition number is that of $\bar{\boldsymbol{A}}$, not its square. The rank decision is a pivot test: the system is refused when $\min_k |R_{1,kk}| \le 10^3\, \varepsilon\, \max_k |R_{1,kk}|$.

**Normal equations.** $\boldsymbol{N} = \sum_k \bar{\boldsymbol{A}}_k^\mathsf{T}\bar{\boldsymbol{A}}_k$, $\boldsymbol{n} = \sum_k \bar{\boldsymbol{A}}_k^\mathsf{T}\bar{\boldsymbol{l}}_k$ and $\bar{\boldsymbol{l}}^\mathsf{T}\bar{\boldsymbol{l}}$ are summed, and $\boldsymbol{N} = \boldsymbol{U}^\mathsf{T}\boldsymbol{U}$ is factorised by Cholesky. Partial systems add, which is the exchange format of the field, but $\kappa(\boldsymbol{N}) = \kappa(\bar{\boldsymbol{A}})^2$ and

$$\bar{\boldsymbol{v}}^\mathsf{T}\bar{\boldsymbol{v}} = \bar{\boldsymbol{l}}^\mathsf{T}\bar{\boldsymbol{l}} - \boldsymbol{n}^\mathsf{T}\hat{\boldsymbol{x}} \tag{1.3}$$

cancels catastrophically when the fit is very good relative to $\|\bar{\boldsymbol{l}}\|$. Where the cancellation exceeds the attainable precision the value is not trusted: the residual sum is recomputed from the residuals in a second pass over the data.

## 1.4 Block-arrow systems and the elimination of local parameters

The parameters split into *common* parameters $\boldsymbol{x}_c$ ($n_c$), seen by every observation, and *local* parameters $\boldsymbol{y}_s$ ($n_{l,s}$) of system $s$ — initial states, instrument calibrations, empirical terms — valid for one arc, one day or the whole system. For the rows of system $s$

$$\bar{\boldsymbol{l}}_s = \bar{\boldsymbol{A}}_s \boldsymbol{x}_c + \bar{\boldsymbol{B}}_s \boldsymbol{y}_s + \bar{\boldsymbol{e}}_s. \tag{1.4}$$

The normal matrix has block-arrow form. The local block of each system — all its arc-, day- and system-scoped parameters together — is eliminated by the Schur complement

$$\boldsymbol{N}_{cc}^{\mathrm{red}} = \boldsymbol{N}_{cc} - \boldsymbol{N}_{lc}^{\mathsf{T}} \boldsymbol{N}_{ll}^{-1} \boldsymbol{N}_{lc}, \qquad \boldsymbol{n}_c^{\mathrm{red}} = \boldsymbol{n}_c - \boldsymbol{N}_{lc}^{\mathsf{T}} \boldsymbol{N}_{ll}^{-1} \boldsymbol{n}_l, \qquad (\bar{\boldsymbol{l}}^{\mathsf{T}}\bar{\boldsymbol{l}})^{\mathrm{red}} = \bar{\boldsymbol{l}}^{\mathsf{T}}\bar{\boldsymbol{l}} - \boldsymbol{n}_l^{\mathsf{T}} \boldsymbol{N}_{ll}^{-1} \boldsymbol{n}_l, \tag{1.5}$$

and the reduced systems of all systems add. On the square-root backend the same reduction is a re-triangularisation with the local columns first,

$$\begin{bmatrix} \boldsymbol{R}_{11} & \boldsymbol{R}_{12} & \boldsymbol{r}_1 \\ & \boldsymbol{R}_{22} & \boldsymbol{r}_2 \\ & & \rho \end{bmatrix},$$

whose trailing rows $[\boldsymbol{R}_{22}\ \boldsymbol{r}_2;\ \boldsymbol{0}^{\mathsf{T}}\ \rho]$ are the reduced contribution; it is accepted only if the local block passes the pivot test. Each system contributes $m_s - n_{l,s}$ degrees of freedom.

Back-substitution recovers

$$\hat{\boldsymbol{y}}_s = \boldsymbol{N}_{ll}^{-1}(\boldsymbol{n}_l - \boldsymbol{N}_{lc}\hat{\boldsymbol{x}}_c), \qquad \boldsymbol{\Sigma}_{\hat{y}\hat{y}} = \hat{\sigma}_0^2 \boldsymbol{N}_{ll}^{-1} + \boldsymbol{G}\boldsymbol{\Sigma}_{\hat{x}\hat{x}}\boldsymbol{G}^{\mathsf{T}}, \quad \boldsymbol{G} = \boldsymbol{N}_{ll}^{-1}\boldsymbol{N}_{lc}. \tag{1.6}$$

Every diagnostic that involves the hat matrix includes the local block. With $\boldsymbol{a} = [\boldsymbol{a}_c; \boldsymbol{b}]$ a row in the full parameter space and $\boldsymbol{z} = \boldsymbol{a}_c - \boldsymbol{G}^{\mathsf{T}}\boldsymbol{b}$,

$$\boldsymbol{a}^{\mathsf{T}}\boldsymbol{Q}_{xx}\boldsymbol{a} = \left\|\boldsymbol{R}_c^{-\mathsf{T}}\boldsymbol{z}\right\|^2 + \left\|\boldsymbol{H}_s\boldsymbol{b}\right\|^2, \qquad \boldsymbol{H}_s^{\mathsf{T}}\boldsymbol{H}_s = \boldsymbol{N}_{ll}^{-1}, \tag{1.7}$$

where $\boldsymbol{R}_c$ is the factor of the reduced common system; the full $u \times u$ cofactor matrix is never formed.

## 1.5 Prior information and regularisation

A prior is a group $k$ of $p_k$ pseudo-observations

$$\boldsymbol{L}_k\boldsymbol{x} = \boldsymbol{x}_{0,k} + \boldsymbol{e}_k, \qquad \mathrm{C}\{\boldsymbol{e}_k\} = \sigma_k^2 \boldsymbol{I}, \tag{1.8}$$

written in the columns of the parameters. In the whitened frame it contributes the rows $\boldsymbol{L}_k/\sigma_k$ with observations $\boldsymbol{x}_{0,k}/\sigma_k$; this is Tikhonov regularisation with $\lambda_k = \sigma_0^2/\sigma_k^2$, $\boldsymbol{N}_\lambda = \boldsymbol{N} + \sum_k \boldsymbol{L}_k^{\mathsf{T}}\boldsymbol{L}_k/\sigma_k^2$. Relative weights inside a group are carried by the rows of $\boldsymbol{L}_k$; $\sigma_k$ is one common factor and can be estimated (§3.7).

Priors on common parameters are stacked into one operator $\boldsymbol{K} = [\boldsymbol{L}_k/\sigma_k]_k$ and applied at the solve: on the square-root backend by appending its rows to the factor, on the normal backend as $\boldsymbol{N} + \boldsymbol{K}^{\mathsf{T}}\boldsymbol{K}$. A prior on local parameters is an observation stream of its own with $\bar{\boldsymbol{A}} = \boldsymbol{0}$, $\bar{\boldsymbol{B}} = \boldsymbol{L}/\sigma$ and $\bar{\boldsymbol{l}} = \boldsymbol{x}_0/\sigma$, so it enters the elimination (1.5) like data. Either way the prior rows count as observations: the redundancy is $n + \sum_k p_k - u$, and the share the data overrule is the redundancy of the group,

$$r_k = p_k - \mathrm{tr}\left(\boldsymbol{L}_k\boldsymbol{N}_\lambda^{-1}\boldsymbol{L}_k^{\mathsf{T}}\right)/\sigma_k^2. \tag{1.9}$$

### 1.5.1 The datum of a free network

A rank defect — a datum that no observation fixes — is removed by a prior on the null space. Let $\boldsymbol{K}_g$ be the rows of group $g$. The group is a *datum* if both the data and every other group are blind to it,

$$\|\boldsymbol{N}\boldsymbol{K}_g^{\mathsf{T}}\|_F \le \varepsilon\,\|\boldsymbol{N}\|_F\|\boldsymbol{K}_g\|_F, \qquad \|\boldsymbol{K}_o\boldsymbol{K}_g^{\mathsf{T}}\|_F \le \varepsilon\,\|\boldsymbol{K}_o\|_F\|\boldsymbol{K}_g\|_F \quad (o \neq g),$$

(on the square-root backend the first test is evaluated on the factor). With $\boldsymbol{K}_d$ the rows of all datum groups and $\boldsymbol{K}_i$ the rest, $\boldsymbol{N}_i = \boldsymbol{N} + \boldsymbol{K}_i^{\mathsf{T}}\boldsymbol{K}_i$. Because the range of the data and the null space are orthogonal, the estimate $\hat{\boldsymbol{x}}$ is the minimum-constraint solution for any datum weight, and its covariance is that of the minimum-constraint problem,

$$\boldsymbol{\Sigma}_{\hat{\boldsymbol{x}}\hat{\boldsymbol{x}}} = \hat{\sigma}_0^2\,\boldsymbol{N}_i^{+}, \tag{1.10}$$

evaluated stably as $\hat{\sigma}_0^2(\boldsymbol{R}_m^{-1}\boldsymbol{R}_m^{-\mathsf{T}} - \boldsymbol{D}_m\boldsymbol{D}_m^{\mathsf{T}})$ with $\boldsymbol{R}_m^{\mathsf{T}}\boldsymbol{R}_m = \boldsymbol{N}_i + c^2\boldsymbol{K}_d^{\mathsf{T}}\boldsymbol{K}_d$, $\boldsymbol{D}_m = \boldsymbol{K}_d^{\mathsf{T}}(\boldsymbol{K}_d\boldsymbol{K}_d^{\mathsf{T}})^{-1}/c$ and a balancing constant $c$. A group only partly in the null space, or one that overlaps another group, is information: its estimate moves with its $\sigma$, and it keeps the covariance $\hat{\sigma}_0^2(\boldsymbol{N} + \boldsymbol{K}^{\mathsf{T}}\boldsymbol{K})^{-1}$.

## 1.6 Nonlinear models: damped Gauss–Newton

For $\boldsymbol{l} = f(\boldsymbol{x}) + \boldsymbol{e}$ the objective, prior included, is

$$F(\boldsymbol{x}) = \big\|\boldsymbol{W}(\boldsymbol{l} - f(\boldsymbol{x}))\big\|^2 + \sum_k \big\|(\boldsymbol{L}_k\boldsymbol{x} - \boldsymbol{x}_{0,k})/\sigma_k\big\|^2. \tag{1.11}$$

At the iterate $\boldsymbol{x}$ the Jacobian $\boldsymbol{J}$ is accumulated once; that accumulation is both the linearisation and the evaluation of $F(\boldsymbol{x})$. The step $\Delta\boldsymbol{x}$ is one least-squares solve with two kinds of pseudo-rows on the common block:

$$\boldsymbol{K}\,\Delta\boldsymbol{x} \approx \boldsymbol{x}_0 - \boldsymbol{K}\boldsymbol{x}, \qquad \sqrt{\lambda d_j}\,\Delta x_j \approx 0, \quad \boldsymbol{d} = \operatorname{diag}(\boldsymbol{J}^{\mathsf{T}}\boldsymbol{P}\boldsymbol{J}). \tag{1.12}$$

The first form makes the iteration converge to the minimiser of (1.11); the second is Marquardt damping of the parameters themselves. A step is accepted only if $F(\boldsymbol{x}+\Delta\boldsymbol{x}) \le F(\boldsymbol{x})$; a rejected step multiplies $\lambda$ by ten (from zero to $10^{-3}$) and is re-solved on the same accumulation, an accepted one divides it by ten (below $10^{-12}$ it is set to zero). The iteration stops when $\max_j |\Delta x_j|/\sigma_{x_j}$ or the relative change of $F$ falls below its tolerance. The covariance is that of the undamped linearised problem at the final point, so it does not depend on the damping history; a run that exhausts its budget returns the best accepted point.

## 1.7 What the estimates mean

$\boldsymbol{\Sigma}_{\hat{\boldsymbol{x}}\hat{\boldsymbol{x}}} = \hat{\sigma}_0^2\boldsymbol{N}^{-1}$ is the *formal* covariance: correct if the functional model is complete, the stochastic model is right and the residuals are white. Chapter 3 makes the second condition true as far as the data allow; Chapter 5 tests the first and the third.

# 2 The stochastic model

## 2.1 Stationary noise per arc

Within one arc of $m$ equally spaced epochs the noise of a scalar stream is modelled as a stationary process with autocovariance $c_\tau$. Its covariance matrix is Toeplitz,

$$\boldsymbol{\Sigma} = \left[c_{|i-j|}\right]_{i,j=1}^{m}, \tag{2.1}$$

and it is positive definite for every $m$ if and only if the spectrum of the process is positive (Bochner). Arcs are independent of each other; non-stationarity across a campaign is carried by one variance factor per arc (§3.6).

## 2.2 Temporal laws

A law is declared by one of the following, each with a level $\sigma$ (a standard deviation):

| law | covariance |
|---|---|
| white | $\boldsymbol{\Sigma} = \operatorname{diag}(\sigma_i^2)$ |
| ARMA$(p,q)$ | $c_\tau$ of the process $\sum_{k=0}^{p} a_k e_{t-k} = \sum_{k=0}^{q} b_k \eta_{t-k}$, $\operatorname{Var}\eta = \sigma^2$ |
| covariance function | $\Sigma_{ij} = \sigma^2 \rho(t_i - t_j)$ (actual time differences) |
| spectrum | $c_\tau = \int_0^{f_\mathrm{N}} S(f)\cos(2\pi f \tau \Delta t)\,\mathrm{d}f$ |
| vector AR$(p)$ | $\boldsymbol{e}_t = \sum_{k=1}^{p} \boldsymbol{A}_k \boldsymbol{e}_{t-k} + \boldsymbol{\eta}_t$, $\mathrm{C}\{\boldsymbol{\eta}\} = \boldsymbol{\Sigma}_\eta$ |

**Spectrum to covariance.** A spectrum is given by ordinates $S_j$ at frequencies $f_j$ and is taken as piecewise linear between them. Its autocovariance is the exact integral; for one segment $[a,b]$ with centre $c$, half-width $h$ and $x = 2\pi\tau h$,

$$\int_a^b S(f)\cos(2\pi f\tau)\,\mathrm{d}f = (S_a+S_b)\,h\cos(2\pi\tau c)\frac{\sin x}{x} - (S_b-S_a)\,h\sin(2\pi\tau c)\frac{\sin x - x\cos x}{x^2}. \tag{2.2}$$

The result is valid at every lag; a non-negative spectrum gives a positive semi-definite covariance of any size.

## 2.3 Whitening an arc

**Dense factor.** $\boldsymbol{W} = \boldsymbol{L}^{-1}$ with $\boldsymbol{L}\boldsymbol{L}^{\mathsf{T}} = \boldsymbol{\Sigma}$, cost $O(m^3)$ per arc. Exact for any positive definite law. An arc is processed in batches of bounded length; a dense law is factorised per batch, so the correlation between batches of a longer arc is neglected.

**Autoregressive filter.** For an AR$(p)$ process with coefficients $\boldsymbol{a} = [1, a_1, \ldots, a_p]$ and innovation standard deviation $s$, whitening is the FIR filter

$$\bar{x}_t = \frac{1}{s}\sum_{k=0}^{p} a_k x_{t-k}, \qquad t > p, \tag{2.3}$$

cost $O(mp)$, with no covariance matrix formed. The first $p$ rows are replaced by the exact stationary start $\boldsymbol{L}_p^{-1}\boldsymbol{x}_{1:p}$, $\boldsymbol{L}_p\boldsymbol{L}_p^{\mathsf{T}} = \operatorname{toeplitz}(c_0,\ldots,c_{p-1})$. The resulting operator is the inverse Cholesky factor of $\boldsymbol{\Sigma}$

itself: both are lower triangular square roots of $\boldsymbol{\Sigma}^{-1}$, which is unique up to sign. Consecutive batches overlap by the filter memory, so batching does not change the result. The AR coefficients of a law given by its autocovariance follow from the Levinson–Durbin recursion (Yule–Walker); a high-order AR can be reduced to an ARMA. A spectrum with narrow features is represented better by the dense factor than by a low-order filter.

## 2.4 Multi-axis streams and the per-epoch accuracy

A stream with $d$ axes per epoch is ordered epoch-major. What the preprocessing knows about the accuracy of epoch $e$ — per-axis standard deviations or a full $d \times d$ covariance $\boldsymbol{C}_e$ — enters as the factor $\boldsymbol{D}_e$ ($\boldsymbol{D}_e = \operatorname{diag}(\sigma_{e,1}, \dots)$, or the lower Cholesky factor of $\boldsymbol{C}_e$). With the temporal law $\boldsymbol{R}_t$ on the epoch grid, the covariance of an arc is

$$\boldsymbol{\Sigma} = \boldsymbol{D}\,(\boldsymbol{R}_t \otimes \boldsymbol{I}_d)\,\boldsymbol{D}^{\mathsf{T}}, \qquad \boldsymbol{D} = \operatorname{blkdiag}(\boldsymbol{D}_e), \tag{2.4}$$

and its whitener factorises: every epoch is first de-scaled by $\boldsymbol{D}_e^{-1}$, then the temporal whitener runs down the epochs of each axis. The order matters: the de-scaled errors are stationary, the raw ones are not when $\boldsymbol{D}_e$ varies. Without per-epoch information $\boldsymbol{D} = \boldsymbol{I}$. For a scalar stream ($d = 1$) the same composition applies with $\boldsymbol{D}_e = \sigma_e$.

Cross-axis correlation *in time* is outside (2.4); it is the vector AR model, fitted by the multichannel Levinson (Whittle) recursion and whitened by the matrix filter $\boldsymbol{w}_t = \boldsymbol{L}_\eta^{-1}\big(\boldsymbol{z}_t - \sum_k \boldsymbol{A}_k \boldsymbol{z}_{t-k}\big)$ with an exact block start.

## 2.5 Where a law can be wrong

A covariance function truncated at a lag shorter than the arc is in general indefinite: its spectrum is the true spectrum convolved with a kernel that has negative side lobes. A law defined by a non-negative spectrum cannot fail this way. An indefinite law is treated as a fault: in accumulation its eigenvalues are floored with a warning; in the variance-component sweep the arc is skipped (§3.8).

# 3 Variance component estimation

## 3.1 The problem

The stochastic model is known up to a finite set of non-negative components entering linearly,

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}) = \sum_{g=1}^{G} \theta_g \boldsymbol{V}_g, \tag{3.1}$$

with known symmetric positive semi-definite matrices $\boldsymbol{V}_g$. The toolbox estimates, jointly in one iteration,

(a) the **spectral shape** of the noise of every pool group of streams (§3.3),
(b) the **level** of every stream,
(c) a **variance factor per arc** (§3.6), and
(d) the **weight of every prior group** declared as estimable (§3.7).

## 3.2 The Helmert estimator and its simplified form

The quadratic forms of the residuals have the expectation (Koch 1999)

$$\mathrm{E}\big[\hat{\boldsymbol{v}}^{\mathsf{T}} \boldsymbol{P}\boldsymbol{V}_g\boldsymbol{P}\hat{\boldsymbol{v}}\big] = \sum_h \theta_h \operatorname{tr}(\boldsymbol{R}\boldsymbol{V}_g\boldsymbol{R}\boldsymbol{V}_h), \qquad \boldsymbol{R} = \boldsymbol{P} - \boldsymbol{P}\boldsymbol{A}\boldsymbol{N}^{-1}\boldsymbol{A}^{\mathsf{T}}\boldsymbol{P}. \tag{3.2}$$

Equating observed and expected forms gives the Helmert system, an unbiased estimator. The simplified estimator drops the coupling between components and iterates

$$\theta_g \leftarrow \theta_g \frac{\Omega_g}{r_g}, \qquad \Omega_g = \theta_g\, \hat{\boldsymbol{v}}^{\mathsf{T}} \boldsymbol{P}\boldsymbol{V}_g\boldsymbol{P}\hat{\boldsymbol{v}}, \qquad r_g = \theta_g \operatorname{tr}(\boldsymbol{R}\boldsymbol{V}_g). \tag{3.3}$$

The ratio $\alpha_g^2 = \Omega_g / r_g$ is the *Helmert factor* of component $g$; the fixed point $\Omega_g = r_g$ ("a weighted residual sum equals its redundancy") is shared with the full estimator. Every component in this chapter is estimated by (3.3); the components differ only in $\boldsymbol{V}_g$.

## 3.3 The spectral law and its components

**Parameterisation.** The law of a pool group is a non-negative spectrum $\boldsymbol{S}$ on $L$ bins $f_j = jd$, $d = 1/(2(L-1))$ cycles per sample, $j = 0, \ldots, L-1$, linear between the bins. Its autocovariance at every lag $\tau = 0, 1, \ldots$ of every arc is the exact integral

$$\boldsymbol{r} = \boldsymbol{K}\boldsymbol{S}, \qquad K_{\tau j} = 2c_j \cos(2\pi f_j \tau)\, \frac{1 - \cos(2\pi\tau d)}{d\,(2\pi\tau)^2}, \qquad K_{0j} = c_j d, \tag{3.4}$$

with $c_j = 2$ inside and $c_j = 1$ at both ends. Column $j$ of $\boldsymbol{K}$ is the autocovariance of the "hat" spectrum of bin $j$, so $\operatorname{toeplitz}(\boldsymbol{K}_{:j})$ is positive semi-definite for every arc length, and the components

$$\boldsymbol{V}_j = \operatorname{toeplitz}(\boldsymbol{K}_{:j}), \qquad \theta_j = S_j \geq 0, \tag{3.5}$$

make (3.1) a sum of positive semi-definite terms. A Helmert factor of a positive semi-definite component cannot be negative, so the law needs no floor and no constraint. The grid sets the spectral resolution, not the lag range: the law is exact at every lag whatever $L$. The law is a *shape*: it is normalised to $r_0 = 1$ after every update, and the level is carried separately. The start is white noise, $\boldsymbol{S} = \boldsymbol{1}$.

**The per-lag statistics.** Write the arc covariance as $\boldsymbol{\Sigma} = \lambda \sum_j S_j \boldsymbol{V}_j$ and let $\boldsymbol{V}^{(n)}$ have ones on the $\pm n$-th diagonals. Per arc and lag the toolbox forms the residual sums and the redundancies

$$\Omega_n = \tilde{\boldsymbol{e}}^{\mathsf{T}} \boldsymbol{V}^{(n)} \tilde{\boldsymbol{e}}, \quad \tilde{\boldsymbol{e}} = \boldsymbol{\Sigma}^{-1} \hat{\boldsymbol{e}}, \qquad s_n = \operatorname{tr}\big(\bar{\boldsymbol{R}} \boldsymbol{V}^{(n)}\big), \tag{3.6}$$

for all lags $n = 0, \ldots, m-1$ of the arc, with $\bar{\boldsymbol{R}}$ the redundancy matrix (3.2) restricted to the arc. Since $\boldsymbol{V}_j = \sum_n K_{nj} \boldsymbol{V}^{(n)}$, the projections on the bins are

$$Q_j = \lambda \sum_n K_{nj} \Omega_n, \qquad T_j = \lambda \sum_n K_{nj} s_n, \tag{3.7}$$

computed for all bins at once by folding the lag sums and one FFT.

## 3.4 The redundancies by random probes

$\Omega_n$ is a sum of products of residuals and is exact. The redundancy needs $\boldsymbol{N}^{-1}$. Per arc, with the whitened designs $\bar{\boldsymbol{A}} = \boldsymbol{W}(\boldsymbol{A} - \boldsymbol{B}\boldsymbol{G})$ and $\bar{\boldsymbol{B}} = \boldsymbol{W}\boldsymbol{B}$, the local block exact and the common block by Hutchinson's estimator,

$$\bar{\boldsymbol{R}} = \boldsymbol{\Sigma}^{-1} - \boldsymbol{R}_t \boldsymbol{R}_t^{\mathsf{T}} - \boldsymbol{K}_l \boldsymbol{K}_l^{\mathsf{T}}, \qquad \boldsymbol{R}_t = \boldsymbol{W}^{\mathsf{T}} \bar{\boldsymbol{A}} \boldsymbol{U}^{-1} \bar{\boldsymbol{Z}}, \qquad \boldsymbol{K}_l = \boldsymbol{W}^{\mathsf{T}} \bar{\boldsymbol{B}} \boldsymbol{L}_l^{-\mathsf{T}}, \tag{3.8}$$

with $\boldsymbol{N} = \boldsymbol{U}^{\mathsf{T}} \boldsymbol{U}$ the reduced common normal matrix (common priors included), $\boldsymbol{N}_{ll} = \boldsymbol{L}_l \boldsymbol{L}_l^{\mathsf{T}}$, and $\bar{\boldsymbol{Z}} = \boldsymbol{Z}/\sqrt{Z}$ a matrix of $Z$ Rademacher probe vectors. Since $\mathrm{E}[\bar{\boldsymbol{Z}}\bar{\boldsymbol{Z}}^{\mathsf{T}}] = \boldsymbol{I}$, $\mathrm{E}[\boldsymbol{R}_t \boldsymbol{R}_t^{\mathsf{T}}]$ is the exact common term. The $u \times Z$ block $\boldsymbol{U}^{-1}\bar{\boldsymbol{Z}}$ is formed once per pass and shared by all arcs (Ellmer 6.5.48); the same probes are used in every pass. Then

$$s_0 = \operatorname{tr} \bar{\boldsymbol{R}}, \qquad s_n = 2 \sum_i \bar{R}_{i+n,i}, \tag{3.9}$$

where the sub-diagonal sums of $\boldsymbol{R}_t \boldsymbol{R}_t^{\mathsf{T}}$ and $\boldsymbol{K}_l \boldsymbol{K}_l^{\mathsf{T}}$ are evaluated for all lags by one zero-padded FFT per column, and those of $\boldsymbol{\Sigma}^{-1}$ by the Levinson recursion and the Gohberg–Semencul representation when $\boldsymbol{\Sigma}$ is Toeplitz (exact columns otherwise). The estimate is unbiased; its spread decreases as $1/\sqrt{Z}$. There is no exact route in the joint estimator; per-observation redundancies for the diagnostics are computed exactly by a different route (§5.3).

## 3.5 One pass

Each pass of the joint estimator

1. accumulates and solves under the current model;
2. sweeps every arc once, whitening it with its current covariance and forming (3.6)–(3.7) and the arc's Helmert factor (§3.6);
3. updates the spectrum of every pool group (streams with an estimated law only),

$$\alpha_j^2 = \frac{\max(\sum_m Q_{jm}, 0)}{\sum_m T_{jm}}, \qquad S_j \leftarrow \alpha_j^2 S_j, \tag{3.10}$$

   summed over the arcs $m$ of the group. A bin whose redundancy $S_j \sum_m T_{jm}$ falls below a threshold keeps its value ($\alpha_j^2 = 1$); an optional moving average smooths $\boldsymbol{S}$; a relative numerical floor of $10^{-10} \max \boldsymbol{S}$ protects the factorisation; then $\boldsymbol{S}$ is renormalised to $r_0 = 1$;
4. updates the arc factors and stream levels (§3.6) and the prior weights (§3.7);
5. records the whiteness of every stream (§5.5).

The iteration stops when

$$\max_j |\alpha_j^2 - 1| \le \varepsilon \qquad \text{and} \qquad \max_m \left| \frac{\sigma^2_{m,\mathrm{new}}}{\sigma^2_{m,\mathrm{old}}} - 1 \right| \le \varepsilon, \tag{3.11}$$

or after the pass budget. After the loop the system is accumulated once more under the final model; that solution is the result.

**Pooling.** Streams of a pool group share one spectrum $\boldsymbol{S}$; every stream keeps its own level. Pooling more streams gives each bin more redundancy per pass; pooling streams of different physics forces one shape on different spectra.

**Policies per stream.** A stream may have its law estimated (shape, level and arc factors), only its level and arc factors (the declared law is kept as the shape), or nothing. An estimated law replaces the declared one from the first pass; the declared level is kept as the multiplier of the estimated components.

## 3.6 Arc factors and stream levels

The covariance of arc $m$ of stream $s$ is

$$\boldsymbol{\Sigma}_m = \sigma^2_{\mathrm{decl},s}\ \sigma^2_m\ \mathrm{toeplitz}(\boldsymbol{K}\boldsymbol{S}_g), \tag{3.12}$$

composed with the arc's per-epoch accuracy (2.4). The Helmert factor of the arc's whole covariance (Ellmer 6.5.27) is

$$\alpha^2_m = \frac{q_m}{t_m}, \qquad q_m = \boldsymbol{r}_m^{\mathsf{T}}\boldsymbol{\Omega}^{(m)}, \quad t_m = \boldsymbol{r}_m^{\mathsf{T}}\boldsymbol{s}^{(m)}, \tag{3.13}$$

with $\boldsymbol{r}_m$ the first column of the arc's temporal covariance, and the updated factor is $\gamma^2_m = \alpha^2_m\sigma^2_m$ (Ellmer 6.5.28). The factors are *absolute*: they weight the solve, so $\sigma^2_m$ is the arc's variance relative to its declaration and $\hat{\sigma}_0$ settles near one. A factor is clamped below at $10^{-8}\bar{\gamma}^2$, with $\bar{\gamma}^2$ the mean of the central half of the stream's factors, so that no arc is silenced without a test having rejected it. A stream without arc factors carries one level, the mean of its arcs' Helmert factors.

The standard deviation of an arc is therefore the product $\sigma_{\mathrm{decl},s}\cdot\sigma_m\cdot\hat{\sigma}_0$.

## 3.7 Weights of prior groups

The weight of a common prior group $k$ (§1.5) is a variance component of its own (Koch and Kusche 2002). From the solution of the previous pass,

$$\Omega_k = \left\|(\boldsymbol{L}_k\hat{\boldsymbol{x}} - \boldsymbol{x}_{0,k})/\sigma_k\right\|^2, \qquad r_k = p_k - \mathrm{tr}\left(\boldsymbol{L}_k\boldsymbol{N}^{-1}\boldsymbol{L}_k^{\mathsf{T}}\right)/\sigma^2_k, \qquad \sigma_k \leftarrow \sigma_k\sqrt{\Omega_k/r_k}, \tag{3.14}$$

so that $\lambda_k = \sigma^2_0/\sigma^2_k$ is the regularisation the data support. A group needs several rows: with one row $r_k \le 1$ and the component is not estimable. A prior on local parameters is an observation stream of its own, and its weight is estimated as that stream's level by (3.3).

## 3.8 Skipped arcs

An arc without data, or whose covariance is not positive definite, is skipped in a pass and keeps its previous factor. A pass in which more than a set share of a stream's arcs is skipped is refused rather than estimated on a fraction of the data.

## 3.9 What the data can resolve

The variance of a component estimated from a redundancy $r_g$ is approximately $2\theta^2_g/r_g$ under normality: a factor with $r_g = 30$ is known to about $\pm 26\,\%$. Few or short arcs cannot carry their own factors; a spectral bin needs enough redundancy per pass, which is what the bin threshold and the choice of the grid control. A spectral law describes stationary noise; a periodic signal or a step in the residuals is a functional model error, which the estimator would absorb into the noise at the price of pessimistic errors (Chapter 5).

# 4 Robust estimation

## 4.1 Influence functions

Robust estimation replaces the weight of an observation by $f(u)\,p$, where $u$ is its standardised residual and $f$ the weight function of an M-estimator. The scale is the centred median absolute deviation, $s = 1.4826\ \text{median}\,|v - \text{median}\,v|$, and $u = v/s$.

| influence | $f(u)$ | default tuning | rejection point |
|---|---|---|---|
| Huber | 1 for $\lvert u\rvert \le c$, $c/\lvert u\rvert$ beyond | $c = 1.345$ | none |
| IGG3 | 1 on $[0,k_0]$, $\dfrac{k_0}{\lvert u\rvert}\Big(\dfrac{k_1-\lvert u\rvert}{k_1-k_0}\Big)^2$ on $(k_0,k_1]$, 0 beyond | $k_0 = 1.5$, $k_1 = 3$ | $k_1$ |
| bisquare | $\big(1-(u/c)^2\big)^2$ for $\lvert u\rvert < c$, 0 beyond | $c = 4.685$ | $c$ |

Huber is monotone and keeps the fraction $c/|u|$ of a gross error's weight; IGG3 and bisquare redescend and reject it. Weights are floored at $10^{-12}$.

## 4.2 Iteratively reweighted least squares on the innovations

Without variance-component estimation, IRLS weights the *law-whitened* residuals (the innovations of the noise process; Yang 1994). Each pass accumulates under the current factors, solves, forms the whitened residuals with the previous factor removed, computes one MAD scale over all data and the factors $f_i$, and applies them after the whitener: the whitened rows are scaled by $\sqrt{f_i}$. The iteration stops when the change of $\hat{\boldsymbol{x}}$ falls below its tolerance relative to the size of the estimate or its formal error. The returned $\hat{\sigma}_0$ and covariance are those of the reweighted problem. For a correlated law a zero innovation weight removes the whitening filter's response to the blunder, not the epoch itself.

## 4.3 Robust weights inside the variance-component estimation

Inside the joint estimator the weights are statements about *observations* (Koch's convention), and they enter the covariance, not the whitened rows.

**The statistic.** The raw residuals $\boldsymbol{v} = \boldsymbol{l} - \boldsymbol{A}\hat{\boldsymbol{x}} - \boldsymbol{B}\hat{\boldsymbol{y}}$ are de-scaled by the declared per-epoch accuracy ($\boldsymbol{D}_e^{-1}\boldsymbol{v}_e$), not by the temporal law and not by the level. For a scalar stream the statistic is the de-scaled residual. For an epoch of $d$ axes it is $\chi$-calibrated,

$$z_e = \operatorname{sign}\Big(\sum_{\text{axes}} v\Big)\,\Phi^{-1}\Big(1 - \tfrac{1}{2}\,Q\big(\tfrac{d}{2}, \tfrac{q_e}{2s^2}\big)\Big), \qquad q_e = \sum_{\text{axes}} v^2, \tag{4.1}$$

with the upper regularised gamma function $Q$, so that a clean $d$-axis epoch is standardised like a clean scalar residual. The scale is one MAD per stream, so the weights do not depend on the level being estimated.

**The nugget.** A factor $f_e$ enters as an additive variance on the diagonal of the temporal law,

$$\boldsymbol{\Sigma} = \boldsymbol{D}\big((\boldsymbol{R}_t + \operatorname{diag}(\boldsymbol{c})) \otimes \boldsymbol{I}_d\big)\boldsymbol{D}^{\mathsf{T}}, \qquad c_e = q_e\,(\boldsymbol{R}_t)_{ee}, \quad q_e = \frac{1}{f_e} - 1. \tag{4.2}$$

The marginal variance of the epoch becomes $(\boldsymbol{R}_t)_{ee}/f_e$, its covariances with the neighbours are unchanged, and $f_e = 0$ deletes the epoch exactly. The arc is whitened by the law's factor followed by an exact rank-$k$ correction over the $k$ down-weighted epochs (Woodbury).

**What uses which factors.** The solution uses all factors. The component update uses only the rejected epochs: a partial factor would bias the level downwards, because the influence function down-weights exactly the epochs with large residuals. The level and the arc factors are divided by the truncation factor of the rejection point $k$,

$$\tau = \frac{F_{\chi^2_{d+2}}(q_k)}{F_{\chi^2_d}(q_k)}, \qquad P(\chi^2_d > q_k) = P(|N(0,1)| > k), \tag{4.3}$$

which is $1 - 2k\varphi(k)/(2\Phi(k) - 1) = 0.9733$ for $d = 1$ and $k = 3$, practically one for bisquare and exactly one for Huber. After the loop the returned solution is the *hard* one: rejected epochs deleted, every partial factor set back to one, and $\hat{\sigma}_0^2$ and the covariance corrected for the deleted rows and for $\tau$. A pass that rejects more than 45 % of the epochs has reached the breakdown point of the estimator.

# 5 Quality assessment

All statistics of this chapter are formed in the whitened frame with a-priori $\sigma_0 = 1$; an "observation" is a whitened row, which for a correlated law is a linear combination of raw epochs. The parameter set is the full one, $u = n_c + \sum_s n_{l,s}$, and the observations include the prior rows, so $n - u$ is the total redundancy. The cofactor matrix is held as the blocks of (1.7).

## 5.1 The order of the verdict

A result is judged in a fixed order, each level assuming the one before it holds:

0. **Integrity.** The redundancy identity $\sum_i r_i = n - u$ holds, and the residual sum of the accumulation agrees with that of a direct pass over the residuals.
1. **Global test.** Is the model as a whole consistent with the declared noise?
2. **Localisation.** Which observations violate it?
3. **Reliability.** What could not have been detected, and how far could it move the estimates?
4. **Parameters.** What is actually known: precision, correlation, conditioning.

## 5.2 Global test

$\hat{\sigma}_0^2/\sigma_0^2$ is tested against $\chi^2_{n-u}/(n-u)$ in the two-sided band $[\chi^2_{0.005}/(n-u),\ \chi^2_{0.995}/(n-u)]$, with the 95 % confidence interval $\left[\hat{\sigma}_0\sqrt{r/\chi^2_{0.975}(r)},\ \hat{\sigma}_0\sqrt{r/\chi^2_{0.025}(r)}\right]$. After variance-component estimation the fixed point $\Omega_g = r_g$ makes the test pass by construction for every estimated group.

## 5.3 Redundancy numbers and the outlier tests

**Redundancy numbers.** $r_i = 1 - \bar{\boldsymbol{a}}_i^{\mathsf{T}} \boldsymbol{Q}_{xx} \bar{\boldsymbol{a}}_i$, evaluated exactly with the local block (1.7). $r_i \to 0$: the observation is fitted whatever its value (no control); $r_i \to 1$: it barely influences the solution.

**Standardised residuals.** For an observation with $r_i > r_{\min}$,

$$w_i = \frac{\bar{v}_i}{\hat{s}\sqrt{r_i}}, \tag{5.1}$$

with $\hat{s} = \hat{\sigma}_0$ (Pope's test, $\sigma_0$ estimated), $\hat{s} = \sigma_0$ (Baarda's test, $\sigma_0$ known) or a robust MAD scale; observations below $r_{\min}$ are uncontrolled and not tested. Under Pope's test $w_i$ follows the $\tau$ distribution, $\tau_r = t_{r-1}\sqrt{r}/\sqrt{r - 1 + t^2_{r-1}}$, bounded by $\sqrt{r}$. With $n$ tests the family-wise level is controlled by Bonferroni, $\alpha' = \alpha/n$.

**Iterative data snooping.** The observation with the largest $|w_i|$ above the critical value is removed, the solution is updated, and the test is repeated with $\alpha$ divided by the remaining number of observations. Removing whitened row $i$ with weight $p$ is a rank-one downdate: with $\boldsymbol{z} = \boldsymbol{a}_c - \boldsymbol{G}^{\mathsf{T}}\boldsymbol{b}$ and $\delta = 1 - p\,\boldsymbol{b}^{\mathsf{T}}\boldsymbol{N}_{ll}^{-1}\boldsymbol{b}$,

$$\boldsymbol{R}_c \leftarrow \mathrm{cholupdate}\big(\boldsymbol{R}_c, \sqrt{p/\delta}\,\boldsymbol{z}, -\big), \qquad \boldsymbol{N}_{ll} \leftarrow \boldsymbol{N}_{ll} - p\,\boldsymbol{b}\boldsymbol{b}^{\mathsf{T}}, \qquad \hat{\boldsymbol{x}} \leftarrow \hat{\boldsymbol{x}} + p\,\bar{v}_i\,\boldsymbol{Q}_{xx}\boldsymbol{a}_i, \tag{5.2}$$

and $\bar{\boldsymbol{v}}^{\mathsf{T}}\bar{\boldsymbol{v}}$ decreases by $p\bar{v}_i^2/r_i$. Observations with low redundancy are not rejected, and the number of rejections is capped; the final model is re-accumulated from the surviving data.

**Robust cross-check.** The raw scores $\rho_i = \bar{v}_i/\sqrt{r_i}$ give a robust scale (the centred MAD). Their ratio to $\hat{\sigma}_0$, and the share $\hat{f}$ of scores beyond 2.5 robust standard deviations (null value $2\Phi(-2.5) = 0.0124$), gauge whether the least-squares scale is inflated by contamination: a contaminated fit inflates $\hat{\sigma}_0$ and masks its own outliers.

## 5.4 Reliability

The minimal detectable bias of observation $i$ is

$$\nabla_{0,i} = \frac{\sigma_i \delta_0}{\sqrt{r_i}}, \qquad 1 - \Phi(k - \delta_0) + \Phi(-k - \delta_0) = 1 - \beta, \quad k = \Phi^{-1}(1 - \alpha'/2), \tag{5.3}$$

the two-sided power solved exactly at the Bonferroni-corrected level ($\delta_0 = 4.13$ for $\alpha' = 10^{-3}$, $\beta = 0.2$). Its effect on the parameters (external reliability) is $\nabla\hat{\boldsymbol{x}} = \boldsymbol{Q}_{xx}\boldsymbol{a}_i\, p_i \nabla_{0,i}$, reported relative to the parameters' standard deviations. Observations with small $r_i$ — short arcs, many local parameters — are where undetected errors hide.

## 5.5 Whiteness and systematic errors

After variance-component estimation the whitened residuals of every stream should be white. The Ljung–Box statistic pools the arcs of a stream: with lag sums $s_k = \sum \bar{e}_i \bar{e}_{i+k}$ and pair counts $N_k = \sum_{\text{arcs}} \max(m_a - k, 0)$,

$$Q = N_0(N_0 + 2) \sum_{k=1}^{K} \frac{\rho_k^2}{N_k} \sim \chi^2_K, \qquad \rho_k = s_k/s_0, \quad K = 20. \tag{5.4}$$

It detects correlation at short lags, periodic terms and excess low-frequency noise; a correlation confined to a single long lag lies outside it. Structure in the residuals against time or another covariate — runs, binned means, periodogram peaks — indicates a *functional* model error. Variance-component estimation cannot repair such an error; it absorbs it into the noise at the price of pessimistic errors and down-weighted arcs.

## 5.6 Parameter-level measures

**Correlation and conditioning.** Correlations follow from $\boldsymbol{Q}_{xx}$. The condition number of $\boldsymbol{N}$ is bounded from below by power iterations on $\boldsymbol{N}$ and $\boldsymbol{Q}_{xx}$; it is a lower bound.

**Contribution of an observation group.** The share of group $g$ in the solution is measured by its redundancy, $r_g/\sum_h r_h$; the per-parameter contribution $\left[\boldsymbol{N}^{-1}\boldsymbol{N}_g\right]_{jj}$, which sums to one over the groups, is estimated with random probes.

**Nested models.** A full model with $u_F$ parameters against a reduced one with $u_R < u_F$ on the same $n$ observations:

$$F = \frac{(\Omega_R - \Omega_F)/(u_F - u_R)}{\Omega_F/(n - u_F)} \sim F(u_F - u_R,\, n - u_F), \tag{5.5}$$

where $u$ counts the common and the eliminated local parameters.

**Confidence ellipses.** For a parameter pair, the ellipse of $\boldsymbol{Q}_{xx}$ restricted to the pair, scaled by $k^2 = 2F_{1-\alpha}(2, r)\hat{\sigma}_0^2$ (or $\chi^2_{1-\alpha}(2)$ when $\sigma_0$ is known).

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